Task 2: Descriptive Statistics and Correlation

Using the traffic dataset, calculate:

2.1 Traffic volume statistics

- Mean
- Median
- Standard deviation
- Variance
- Range

Interpret what these statistics tell you about traffic volume and its variability.

Refer to the Power Bi Report:

What the cards show

Measure	Value
Row Count	40,575 hours
Mean	3,290.65
Median	3,427
Std Dev (Sample)	1,984.77
Variance (Sample)	3,939,324
Min / Max	0 / 7,280
Range	7,280

1. **Typical level.** Traffic averages about 3,291 vehicles per hour, with a median of 3,427. The median is slightly above the mean, so the lower values pull the average down a little.
2. **High variability.** The standard deviation (1,985) is about 60% of the mean, and the variance is large (about 3.9 million). A typical hour sits roughly 2,000 vehicles away from the average, so volume is spread widely rather than clustered around the mean.
3. **Wide range.** Volume ranges from 0 to 7,280, so the highest hour is more than twice the mean. The minimum of 0 shows there are hours with no recorded traffic.

What the histogram shows

With a bin size of 500, the histogram has two clear peaks:

Bin	Hours
0	4,749

Bin	Hours
500	4,270
4000	3,341
4500	4,800
5000	3,601
7000	55

- **Two peaks:** One is at the low end (bins 0 and 500) and another is around 4,500 to 4,999 (4,800 hours, the tallest bin).
- **A dip between them:** Counts fall to 1,514 hours in the 1500 bin before rising again.
- **A thin upper tail:** Very few hours reach 6,500 or above (935 in the 6500 bin and 55 in the 7000 bin).

What this means

- **The mean describes few hours.** It falls at 3,291, in the lower part of the middle between the two peaks, where the bins hold fewer hours than at either peak. Most hours are either quite low or quite high, which explains the large standard deviation.
- **Traffic volume is not evenly distributed.** The report supports describing traffic as having two distinct levels, a low-traffic group and a high-traffic group, rather than a single typical level.
- **Limit of the report:** The cards and histogram show how volume is spread but not why. They do not show what causes the two levels, such as time of day or day of the week. To explain it, you would add a chart of average volume by hour or by weekday to the report.

2.2 Correlation analysis

Calculate the correlation coefficient between:

- Temperature
- Traffic volume

Then:

- Interpret the **direction** of the relationship.
- Interpret the **strength** of the relationship.

Explain why correlation does not necessarily imply causation

Refer to the Power Bi Report:

Direction of the relationship

- **Positive.** The correlation is r = 0.1393, which is above zero, so warmer hours tend to go with slightly higher traffic and colder hours with slightly lower traffic.
- The report supports this in two places. The trend line on the scatter chart slopes gently upward, and average traffic is about 2,500 vehicles per hour in the coldest band (−30 to −21°C) against about 4,800 in the warmest (30°C and above).
- The pattern is not perfectly steady. The 10 to 19°C band is a little below the 0 to 9°C band, which is one reason the correlation is weak.

Strength of the relationship

- **Weak.** An r of 0.139 is close to zero (0 means no linear relationship, 1 means a perfect one).
- R^2 = 0.0194, so temperature explains only about 2% of the variation in hourly traffic. The other 98% comes from other factors.
- The scatter chart shows the same thing: at almost any temperature, traffic ranges from near 0 to about 7,000 vehicles per hour. Knowing the temperature tells you very little about how busy the road will be in a given hour.
- This fits the Task 2.1 result, where the standard deviation (1,985) was large. Temperature cannot account for most of that spread.
- With 40,565 hours of data, even a small correlation can be statistically detectable, but that does not make it important in practice.

Why correlation does not imply causation

A correlation shows that two variables move together. It does not show that one causes the other, for three reasons:

1. **A third factor may drive both.** Temperature and traffic both change with time of day and time of year. For example, afternoons are generally warmer than early mornings, and traffic is also higher in the afternoon. They could then move together even if temperature has no effect on driving. The report does not test this, so it is a possible explanation, not a finding.
2. **The direction is unclear.** The correlation alone cannot show whether temperature affects traffic or something else is behind both. Traffic does not change the weather, but without extra evidence the numbers cannot rule out other explanations.
3. **Coincidence and weak effects.** A small correlation can appear in large datasets by chance or through unrelated patterns. The warmest band (30°C and above) also contains far fewer hours than the others, so its high average is less reliable.

To establish causation, you would need to hold other factors constant (for example, compare the same hour of day and the same season at different temperatures) or run a controlled experiment.

Power BI Report Screenshot reference:

2.1 Traffic volume statistics

40575	3290.65	3427.00	1984.77
Row Count	Mean Traffic	Median Traffic	Std Dev (Sample)
3939323.50	0.00	7280.00	7280.00
Variance (Sample)	Min Traffic	Max Traffic	Traffic Range

Traffic volume: summary statistics

- Based on 40,575 hourly records, average traffic is about 3,291 vehicles per hour. The median is higher at 3,427.
- The standard deviation is 1,985, about 60% of the mean, so traffic varies a lot from hour to hour.
- Volume ranges from 0 to 7,280 vehicles per hour, so the busiest hour is more than twice the average.
- The average alone is not a reliable guide to what traffic is like at any given hour.

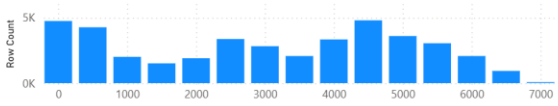
2.2 Correlation analysis

0.1393	0.0194	40565
Correlation (r)	R Squared	Hours Used

Temperature vs traffic: correlation

- The correlation is r = 0.139, a weak positive relationship: warmer hours tend to have slightly higher traffic.
- R^2 = 0.019, so temperature explains only about 2% of the variation in traffic.
- The result uses 40,565 hours. Ten rows with an impossible temperature reading (0 Kelvin) were excluded.

Row Count by traffic_volume (bins)



Distribution of hourly traffic volume

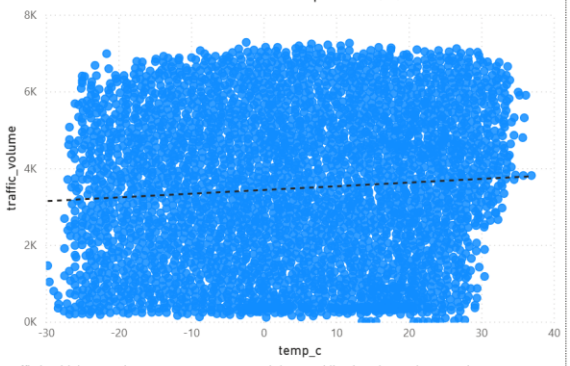
- Hours cluster at two levels: a low group (0 to about 1,000 vehicles) and a high group (about 4,000 to 5,500).
- Between them, hours are less common, with the fewest in the 1,500 to 2,000 range.
- The mean (about 3,291) sits between the two groups, so it describes relatively few actual hours.
- Very few hours exceed 6,500 vehicles.

Mean Traffic by temp_c (bins)



Average traffic generally rises with temperature, from about 2,500 in the coldest band to about 4,800 in the warmest, but not smoothly.

Traffic volume vs temperature (°C)



Traffic is widely spread at every temperature, and the trend line is only gently upward. Temperature alone tells you little about how busy the road is.